

RHEA ADDEPALLI

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AI/ML Engineer with hands-on experience in building and deploying AI-driven systems using Generative AI, LLMs, and agent-based architectures. Skilled in developing RAG pipelines, multi-stage workflows, and AI-assisted solutions using Python, PyTorch, and TensorFlow. Experienced in prompt engineering, embeddings, NLP, and model evaluation, with a strong focus on real-world problem solving and scalable system design.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering in AIML

- Alliance University, Bengaluru | Year of Graduation: 2026 | CGPA: **9.1** | Backlogs : None

12th Grade (HSC)

- Sri Chaitanya Junior College, Andhra Pradesh | State Board | Year: 2022 | Percentage: **98.3%**

10th Grade (SSLC)

- Sri Chaitanya School, Andhra Pradesh | State Board | Year: 2020 | Percentage: **97.5%**

Skills

Languages: Python, Java, HTML

AIML Tools : PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, YOLO, OCR, Hugging Face.

GenAI & LLM: LLMs, RAG, Prompt Engineering, Agentic AI, Multimodal AI, Conversational AI, Gemini API

Development & Collaboration Tools: SQL, RDBMS, Git, GitHub, Flask, REST API, JSON, CI/CD Pipelines.

AI Concepts : Machine Learning, Deep Learning, Reinforcement Learning, Neural Networks, XAI, GenAI.

Practices: Experiment tracking, benchmarking, and clear technical documentation

AI Tools & Platforms: ChatGPT, Microsoft Copilot, Claude, Gemini API

Projects

Memory-Augmented Conversational RAG System with Agent-Based Reasoning– [Python, FAISS, LLMs, NLP, Agents, RAPTOR, RAG, JSON, REST API]

- Built a conversational RAG-based document QA system with memory for multi-turn interactions
- Integrated open-source LLMs for prompt-based generation and response synthesis
- Designed FAISS semantic retrieval using embeddings, reranking, and RAPTOR for context understanding
- Developed a LangGraph-style stateful pipeline with workflow orchestration, conditional routing, and multi-stage agent reasoning
- Implemented ReAct-based agent workflow for multi-step reasoning and AI-driven automation
- Developed REST API services using Flask for scalable AI application deployment and integration
- Built a production-grade evaluation and benchmarking pipeline with grounding, Recall@K, confidence scoring, hallucination detection, and multi-run tracking

XAI Bird Classification - [LIME, CNN, NumPy, TensorFlow, FastAPI, Angular, CI/CD, Fine-Tuning]

- Fine-tuned MobileNetV2 and benchmarked against MobileNetV3 and EfficientNet (~98% accuracy)
- Applied LIME for interpretable predictions and model explainability
- Built and deployed a real-time Angular + FastAPI application with TFLite optimization
- Implemented CI/CD pipelines and evaluated model using accuracy, confusion matrix, and latency.
- Deployed model on Render - <https://xai-bird-classification-using-cnn.onrender.com>

GenAI DocBot [Python, Gemini API, PHP, JavaScript, PyMuPDF, FAISS, Railway]

- Built a scalable PDF summarization system using Gemini API with map-reduce chunking (300+ pages)
- Implemented parallel processing (ThreadPoolExecutor) for faster inference
- Added MD5-based caching to avoid redundant API calls
- Designed user privacy controls for document storage and deletion

Certifications

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1. Cloud Infrastructure Certified Generative AI Professional - Oracle | 2025 - [Link](#)
 2. Ai Foundations Associative - Oracle | 2025 - [Link](#)
 3. Introduction to Agile Development and Scrum - IBM(Coursera) | 2025 - [Link](#)
 4. Data Structures - UC San Deigo(Coursera) | 2024 - [Link](#)
 5. Relational Database Design – Coursera | 2023 - [Link](#)